

4.5 Applied Optimization

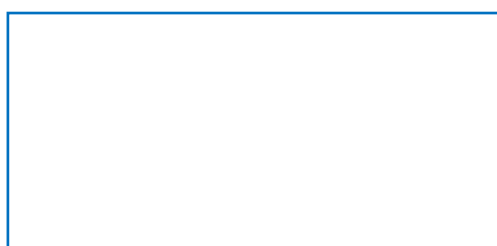
What are the dimensions of a rectangle with fixed perimeter having *maximum area*? What are the dimensions for the *least expensive* cylindrical can of a given volume? How many items should be produced for the *most profitable* production run? Each of these questions asks for the best, or optimal, value of a given function. In this section we use derivatives to solve a variety of optimization problems in mathematics, physics, economics, and business.

◆ Solving Applied Optimization Problems

1. **Read the problem.** Read the problem until you understand it. What is given? What is the unknown quantity to be optimized?
2. **Draw a picture.** Label any part that may be important to the problem.
3. **Introduce variables.** List every relation in the picture and in the problem as an equation or algebraic expression, and identify the unknown variable.
4. **Write an equation for the unknown quantity.** If you can, express the unknown as a function of a single variable or in two equations in two unknowns. This may require considerable manipulation.
5. **Test the critical points and endpoints in the domain of the unknown.** Use what you know about the shape of the function's graph. Use the first and second derivatives to identify and classify the function's critical points.

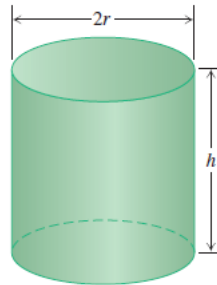
① Maximum Area

Example 1 A field biologist want to enclose a rectangular study plot. She has 1600 ft of fencing. Using this fencing, what are the dimensions of the study plot that will have the largest area?



② Minimizing Material

Example 2 You have been asked to design a one-liter can shaped like a right circular cylinder. What dimensions will use the least material?



Example 3 A rectangle is to be inscribed in a semicircle of radius 2. What is the largest area the rectangle can have, and what are its dimensions?

